

Aravindhan TV

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SKILLS

Languages:	C, Embedded C, Assembly (8051), Python, Bash
Tools/Platforms:	STM32 Cube, MBED platform, Keil, Kicad, Arduino IDE, Creo, Proteus, Cadence - NC Sim
Technical Skills:	Digital Circuits, Analog circuits, Soldering, PCB design, Circuit design, Sensor integration, Communication protocol (UART, SPI, I2C), Wireless communication (WiFi, Bluetooth), Operating system (linux), Development boards - Raspberry pi, STM32 Nucleo F401RE, Arduino - Uno/Nano, ESP32/ESP8266
Soft Skills:	Problem-Solving Skills, Team Player, Project Management, Adaptability

TRAINING

PantechElearning - Edutech company

Jun '25

PCB Design Intern

- **About:** Learned about the PCB designing using the tool Kicad(an open source pcb designing tool). In this training I had learnt how to design a schematic and conversion of the schematic into a PCB, the routing of wires. Created the custom footprint for the component. We had designed a pcb of ESP32 from scratch till 3D view and generated the Gerber file.
- **Tech stacks used:** Kicad, circuit designing.

PROJECTS

Edge-Intelligent Industrial Safety Monitoring System:

Dec '25 - Designed and developed an industrial safety and condition monitoring system with local edge intelligence.

- Implemented vibration using MPU6040 (IMU) and temperature using DS18B20, anomaly detection using derived signal features such as delta, RMS, EWMA, and residual analysis.
- Developed multi-stage decision logic with confirmation counters to reduce false alarms and improve reliability.
- Implemented non-blocking local safety actions using LED and buzzer with critical-event latching.
- Integrated ESP32-CAM for visual inspection, enabling event-triggered image capture during critical conditions for enhanced situational awareness.
- Designed a real-time web dashboard using Node.js, SQLite, and WebSockets for event visualization, image logging, and historical analysis.
- Built a modular, scalable system architecture suitable for industrial monitoring, visual inspection, and predictive maintenance applications.

Tech Stack: ESP32-S3, ESP32-CAM, Embedded C/C++, MPU6050, DS18B20, Node.js, SQLite, WebSockets

Github repository:

RF-IMU Data Bridge:

Nov '25 - Designed and developed a low-cost wireless sensor telemetry system for real-time motion data transmission using sub-GHz

- RF communication.
- Implemented inertial sensing using MPU6050 (3-axis accelerometer and gyroscope) interfaced via I2C for motion and orientation data acquisition.
- Designed custom data packet framing and transmission logic over 433 MHz RF (STX/SRX) modules, enabling reliable point-to-point wireless communication without Wi-Fi or Bluetooth.
- Developed transmitter and receiver firmware on ESP32, handling sensor initialization, data formatting, RF transmission, reception, and decoding.
- Implemented timing control and transmission delays to improve RF stability and reduce packet collision and noise effects.
- Integrated a 16x2 I2C LCD at the receiver node for real-time visualization of received IMU parameters.
- Validated system performance through continuous real-time data streaming and hardware-level debugging of I2C and RF interfaces.

Tech Stack: ESP32, MPU6050, 433 MHz RF (STX/SRX), Embedded C/C++, I2C, Arduino IDE.

Project CAPTURE | Autonomous inspection robot:

Mar'25 - Developed and autonomous inspection robot for pipeline inspection for oil and gas industries which can move autonomously without any manual control

- Designed the structure of the robot and fabricated the body using MDF(Medium density fiber) and metal sheets

- Implemented this project with a raspberry pi 4 b+ and Arduino uno. Raspberry pi handles all the sensors(BMP280, MQ2, MLX90640) and develop the program with Python on RaspbianOS CLI environment). Arduino UNO handles the movement and navigation of the bot including the power management from the battery to the motors, controlling the motor driver.

- This project is made within 24 hour in an hackathon

Tech Stack: Raspberry Pi, Python, Arduino, Flask server, WIFI.

CERTIFICATES

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| • Embedded system Essentials - ARM education, Edx (Link) | Jul '25 |
| • Computer Architecture and Organization - IIT Kharagpur, NPTEL (Link) | Nov '25 |

ACHIEVEMENTS

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| • Winner
Fit-Fusion, Cognizance, IIT Roorkee | Mar' 25 |
| • First Runner up
R2VC workshop, IIT Roorkee - 2025 | Mar' 25 |
| • Shortlisted by Ministry of Education's Innovation cell:
Sep '24 IDE Bootcamp, Amity Jaipur - 2024 | |
| • Winner
Ideathon Bootcamp, LPU Punjab - 2023 | Oct '23 |

EDUCATION

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|--|-----------------------------------|
| • Lovely Professional University
Bachelor of Technology - Computer Science and Engineering; CGPA: 7.53
Aug '23 | Punjab, India
Since |
| • Carmel Matriculation Higher secondary school
Nadu
High School | Erode, Tamil
Jun '21 - Mar '23 |
| • Carmel Matriculation Higher secondary school
Nadu
Matriculation | Erode, Tamil
Mar '21 |